

analysis_scripts

Analysis Scripts (Quick Reference)

All analysis scripts are in `analysis/`. They read simulation data from:

- `patt1d_inputs/<filename>/seed.in`
- `patt1d_outputs/<filename>/...`

How To Run

From project root:

```
python analysis/<script_name>.py -h
```

Typical usage pattern:

```
python analysis/<script_name>.py -f <sim_filename> [other flags]
```

Script List

Script	What it does	Typical r
<code>analyze_delay_times_1D_BEC.py</code>	Finds observed delay times from first-harmonic peaks and compares with analytic scaling.	<code>python a</code> <code><sim_fi</code>
<code>calc_area_of_ft_data.py</code>	Computes Fourier-peak areas over selected frame ranges.	<code>python a</code> <code>0 -s 0 -</code>
<code>calc_legett_at_t.py</code>	Computes Leggett-style quantity at a selected time/frame.	<code>python a</code> <code>-i 20</code>
<code>calc_oscill_period_at_t.py</code>	Estimates oscillation period	<code>python a</code> <code>-t 1e8 -</code>

Script	What it does	Typical run
	at a selected time/frame.	
<code>compare_sqrt_vs_mean.py</code>	Demonstrates difference between $\langle \sqrt{S} \rangle$ and $\sqrt{\langle S \rangle}$ on a toy profile.	python 3
<code>export_s_statistics.py</code>	Exports spatial profiles of $\langle \sqrt{\text{abs}(S)} \rangle$ and $\sqrt{\langle \text{abs}(S) \rangle}$ after analytic delay t_0 .	python 3 25
<code>plot_analytic_and_observed_omega_gamma.py</code>	Plots analytic vs observed frequency/growth diagnostics vs p_0 .	python 3 <code><sim_fi</code>
<code>plot_density_evolution_at_t.py</code>	Plots density `	psi
<code>plot_density_evolution_at_x.py</code>	Plots density `	psi
<code>plot_density_first_harmonic_max_vs_p0.py</code>	For each density file, finds max first-harmonic amplitude vs time and plots it vs p_0 .	python 3 <code><sim_fi</code>
<code>plot_first_harmonic_amp_vs_time_density.py</code>	Plots density first-harmonic amplitude vs time for one frame index.	python 3 f <code><sim_</code>
<code>plot_first_harmonic_amp_vs_time_intensity.py</code>	Plots intensity first-harmonic amplitude vs time for one frame index.	python 3 -f <code><sim_</code>
<code>plot_ft_at_t.py</code>	Plots spatial Fourier transform at a selected time.	python 3 20

Script	What it does	Typical results
<code>plot_ft_at_x.py</code>	Plots temporal Fourier transform at a selected spatial position.	<code>python plot_ft_at_x.py -f <sim_file> -x 0 -i 1</code>
<code>plot_intensity_first_harmonic_max_vs_p0.py</code>	Intensity analogue of max first-harmonic vs p_0 .	<code>python plot_intensity_first_harmonic_max_vs_p0.py -f <sim_file></code>
<code>plot_s_evolution_at_x.py</code>	Plots intensity $S(t)$ at one spatial position.	<code>python plot_s_evolution_at_x.py -x 0 -i 1</code>
<code>plot_s_ft_at_t.py</code>	Plots spatial Fourier transform of intensity at one time.	<code>python plot_s_ft_at_t.py -t 1e8</code>
<code>plot_spatial_analysis_graphs_changing_p0.py</code>	Multi-panel spatial statistics vs p_0 (density).	<code>python plot_spatial_analysis_graphs_changing_p0.py -f <sim_file></code>
<code>plot_spatial_ft_graphs_changing_p0.py</code>	Multi-panel spatial Fourier metrics vs p_0 .	<code>python plot_spatial_ft_graphs_changing_p0.py -f <sim_file></code>
<code>plot_spatial_mod_depth_vs_p0.py</code>	Spatial modulation depth / first-harmonic metrics vs p_0 (density).	<code>python plot_spatial_mod_depth_vs_p0.py -f <sim_file></code>
<code>plot_spatial_mod_depth_vs_p0_intensity.py</code>	Spatial first-harmonic metric vs p_0 (intensity).	<code>python plot_spatial_mod_depth_vs_p0_intensity.py -f <sim_file></code>
<code>plot_temporal_analysis_graphs_changing_p0_at_x.py</code>	Temporal statistics vs p_0 at fixed x (density).	<code>python plot_temporal_analysis_graphs_changing_p0_at_x.py -f <sim_file></code>
<code>plot_temporal_delay_vs_p0_at_x_using_fourier_amp.py</code>	Delay time vs p_0 from first-harmonic temporal peaks + analytic overlay.	<code>python plot_temporal_delay_vs_p0_at_x_using_fourier_amp.py -f <sim_file></code>
<code>plot_temporal_ft_graphs_changing_p0_at_x.py</code>	Multi-panel temporal Fourier	<code>python plot_temporal_ft_graphs_changing_p0_at_x.py -f <sim_file></code>

Script	What it does	Typical run
	metrics vs p_0 at fixed x .	
<code>plot_temporal_mod_depth_vs_p0_at_x.py</code>	Temporal modulation depth vs p_0 at fixed x (density).	<code>python <sim_file></code>
<code>reindex_p0_outputs.py</code>	Renames <code>psi*.out</code> and <code>s*.out</code> so indices increase with p_0 .	<code>python <script> --dry-run</code>

Notes

- Use `-h` for full options on any script.
- Many scripts open interactive windows; some also save files.
- Most scripts assume one simulation folder name passed via `-f`.